

ARTICLE

Tree species size class patterns portend compositional shifts and low resilience in managed northern hardwood forests

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Abstract

Declining temperate forest biodiversity is a global challenge to future sustainability and resilience. Canopy tree diversity has declined in northern hardwood forests (NHF) of the Great Lakes region since European colonization, with current tree regeneration patterns suggesting this trend may continue. However, regional (10^6 ha) patterns and drivers are underexplored. We used data from 141 managed NHF stands, Michigan, USA, to examine diversity indices across tree size classes (seedling, sapling, and canopy); stand-level drivers of regeneration (seedling and sapling) diversity and individual species density; and patterns among species in relative abundances across size classes. Diversity was similar across size classes (mean asymptotic species richness of 6–7 species and <3 effective common species); however, some species contributing to sapling diversity are unlikely to contribute to future canopy diversity (*Fraxinus americana* and *Fagus grandifolia*, insect/pathogen limited; *Ostrya virginiana*, small maximum size). For the 11 most common species, conspecific canopy density (+, seedlings, significant for 10 species) and stand basal area (–, saplings, 3 species) most consistently associated with density; deer use, site quality, and substrate were less predictive. Patterns of relative density by size class among 18 species present on >10% of sites were consistent with species life history strategy and/or establishment/recruitment barriers. For seven species (including *Tsuga canadensis*, *Betula alleghaniensis*), relative density was lowest in the seedling layer, suggesting seedling establishment substrate and/or early shade mortality limitations. For two species (*Acer saccharum*, *Quercus rubra*), reduced sapling relative density compared with seedling or canopy strata suggests a sapling recruitment bottleneck from deer browsing/shade, whereas six species (e.g., *F. grandifolia*, *O. virginiana*, *Pinus strobus*) had highest relative density for saplings, suggesting sapling to canopy recruitment limitations. Lastly, seedlings were relatively most dense for *Acer rubrum* and *Prunus* spp., indicating disproportionally high seedling establishment, if not sapling or canopy recruitment, and for *F. americana*, consistent with recent pest-related canopy mortality. Collectively our analyses suggest sustained low diversity and shifting composition are driven by limited local seed availability, limited seedling establishment

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substrate, deep shading, and abundant deer. Management aimed at overcoming these limitations may be necessary to promote future NHF resilience.

KEYWORDS

Hill numbers, northern hardwood forest, regeneration, seedling–sapling–canopy, single-tree selection, species diversity

INTRODUCTION

Tree diversity is a main determinant of forest ecosystem resilience, which is the capacity of forests to return to their original state following disturbance (Thompson et al., 2009). However, in some regions of the northern temperate forests of eastern North America, current tree regeneration patterns suggest declining tree species diversity and changing species composition over the coming decades (Miller & McGill, 2019; Ramirez et al., 2018; Vickers et al., 2019). Compromised diversity may put forest functioning at risk and challenge forest resilience, especially given changing and novel disturbance regimes. Notable stressors that are changing include forest fire activity (Flannigan et al., 2000), forest management/harvesting paradigms (Kern et al., 2014), and herbivore/browsing pressure (Matonis et al., 2011), plus novel stressors including invasive pests and pathogens (Dukes et al., 2009; Ramsfield et al., 2016) and climate change. Some of these stressors directly contribute to declining tree diversity and shifting composition (e.g., forest harvesting since European colonization; Schulte et al., 2007), suggesting current disturbance regimes represent a self-reinforcing cycle of declining diversity and decreasing resilience. Characterizing tree regeneration species diversity at the stand level (i.e., the unit of forest management) and factors associated with diversity over broad geographic extents is key for forecasting future forest resilience and for altering management to bolster resilience in increasingly volatile environments.

Northern hardwood forests (NHF), comprising 20 million ha of the northern temperate forest biome (Braun, 1950; Leak, 1987; Rowe, 1972), are currently dominated by ecologically and economically valuable sugar maple (*Acer saccharum* Marsh) (Duval et al., 2014; Linehan & Jacobson, 2005) with variable admixtures of ~20 other broad-leaved deciduous and evergreen conifer species (Braun, 1950). Concerningly, NHF canopy diversity has declined since European colonization (Schulte et al., 2007) and is lower in managed stands than in unmanaged stands (Crow et al., 2002; Neuendorff et al., 2007; Powers & Nagel, 2009). Current regeneration (i.e., seedling/sapling class) in some areas has low diversity, with low density of sugar maple and some other species in

the overstory, suggesting future compositional shifts in the canopy (Elenitsky et al., 2020; Hupperts et al., 2020; Leak, 2006; Matonis et al., 2011; Neuendorff et al., 2007; Powers & Nagel, 2009). These changes could inhibit long-term forest capacity to provide multiple ecosystem goods and services, including timber, carbon sequestration, and wildlife habitat (Walters et al., 2022). Addressing challenges to securing diverse tree regeneration is becoming increasingly important, given predicted declines in NHF ecosystem functioning from climate change (Rogers et al., 2017) and continued invasive pest and pathogen establishment and spread (Chapman et al., 2017; Liebhold et al., 2013; Panzavolta et al., 2021).

Multiple factors are known to influence the composition and diversity of tree regeneration in NHFs. In the Great Lakes region where NHFs are common, postglacial landforms drive spatial variation in nutrient availability and soil water-holding capacity (site quality), and related forest type distributions, from xeric jack pine/black oak dominated coarse-textured outwash soils, to mesic northern hardwood-dominated finer textured moraine soils (Baribault et al., 2010; Zak et al., 1986, 1989). Interacting with site quality is a dramatic gradient of annual snowfall (e.g., differences of a few meters for sites <100 km apart) driven by the Great Lakes (i.e., lake-effect), which further influences forest and tree species distributions, particularly sugar maple given its association with high snowfall areas (Henne et al., 2007). Over the narrower range of higher site qualities supporting NHFs, tree canopy and regeneration composition vary (Burger & Kotar, 2003; Elenitsky et al., 2020). Compositional data supporting habitat classification development for the region (i.e., frequency of occurrence of saplings, Burger & Kotar, 2003) suggest that lower site quality stands supporting NHFs tend to have more species; however, the influence of site on species composition may be overriden by other factors, such as deer browsing or management (Bannon et al., 2015; Matonis et al., 2011).

Several disturbance agents may interact with site quality to drive changes in tree composition and diversity in NHFs. Invasive pests and pathogen epidemics have recently greatly decreased the likelihood of canopy recruitment for American beech (*Fagus grandifolia*),

white ash (*Fraxinus americana*), and American elm (*Ulmus americana*) (Forrester et al., 2003; Klooster et al., 2014; Nuckolls et al., 2009; Parker & Leopold, 1983). Impacting a wider range of tree species, invasive non-native earthworms may change tree regeneration composition by altering nutrient cycling, changing forest floor structure, and reducing understory plant density (Bohlen et al., 2004; Hale et al., 2006; Holdsworth et al., 2007, 2008; Resner et al., 2015). Abundant white-tailed deer (*Odocoileus virginianus*) populations, which have increased dramatically since the early 1900s (MDNR, 2016), can shift species composition through selective browsing and depress overall regeneration density (Bradshaw & Waller, 2016; Côté et al., 2004), including canopy-dominant sugar maple (Curtis & Rushmore, 1958; Henry et al., 2021; Horsley et al., 2003; Kain et al., 2011; Matonis et al., 2011; White, 2012). Deer browsing interacts with stand-level variables to influence tree regeneration density and composition, including site quality (Randall & Walters, 2011) and light availability/gap size (Sage et al., 2003; Walters et al., 2016). Since deer browsing is vertically limited to a maximum of ~2 m (Walters, Farinosi, & Willis, 2020), regeneration which has vertically escaped the deer browsing zone may be particularly indicative of future canopy structure and composition (Harris et al., 2022). Capturing variability in biotic drivers, such as invasive pests and pathogens or local deer populations, is key to fully characterizing trends in tree regeneration diversity and distribution, which is necessary for identifying particularly low-diversity, resilience-challenged regions in need of alternative management intervention.

Forest harvesting regimes have contributed to past changes in composition and declining diversity for both canopy (Crow et al., 2002; Neuendorff et al., 2007; Powers & Nagel, 2009; Schulte et al., 2007) and regeneration layers (Powers & Nagel, 2009). The single-tree selection (STS) silvicultural system was introduced in the late 1950s (Arbogast, 1957) and has since been the recommended standard for NHF management in the Great Lakes region (Kern et al., 2014), though implementation of selection silviculture often varies widely and far from the original guidelines (Pond et al., 2014). Under this system, periodic partial harvests every 10–20 years remove select trees, resulting in small, dispersed harvest gaps. These gaps theoretically promote the regeneration of economically desirable shade-tolerant species, such as sugar maple. However, this system may be leading to declines in tree diversity via unintentional or intentional local (sub-stand to stand scale) elimination of species (e.g., weeding species of low economic value); this ultimately leads to elimination of canopy (i.e., mature tree) seed sources (Willis et al., 2016). Local elimination of

seed sources may be particularly limiting for species with shorter dispersal distances (McEuen & Curran, 2004). Further, the low light regeneration environments resulting from small STS harvest gaps can decrease regeneration and canopy recruitment of less shade-tolerant species (Crow et al., 2002; Niese & Strong, 1992; Walters et al., 2016). In addition, small-seeded species, such as *Betula alleghaniensis* and *Tsuga canadensis*, may face additional seedling establishment constraints via high coverage of germination-establishment inhibiting broad-leaved leaf litter and low coverage of favorable coarse wood (from periodic harvests removing large trees before they die) and bare mineral soil (BMS, via minimal disturbance from winter logging and low-impact harvesting equipment) germination-establishment substrates (Bolton & D'Amato, 2011; Marx & Walters, 2008; Willis et al., 2016).

While STS management may contribute to low diversity in NHFs, alternative management systems can play a role in improving future diversity. Alternatives to STS, including more intense harvesting regimes, are being explored to improve regeneration outcomes (Hupperts et al., 2020; Rogers et al., 2021; Walters, Roloff, et al., 2020), including increasing diversity. For example, larger gaps or other more intense harvests (e.g., shelterwood harvests) or variable harvest intensity within stands may promote greater diversity via admixtures of both shade-intolerant and -tolerant tree species (Niese & Strong, 1992; Walters et al., 2016). However, despite the theoretical appeal of more intense or mixed intensity harvest regimes, evidence to date is mixed, with some studies suggesting more intense or variable intensity harvests will not increase regeneration diversity in the long term (Kern et al., 2013; Knapp et al., 2021), whereas others do (Niese & Strong, 1992; Walters et al., 2016). Accurately characterizing current regeneration composition and diversity patterns and associated driving factors associated with management is necessary to guide development and implementation of future management of NHFs, whether by continued use of STS or by alternative silvicultural strategies. In addition, future management is likely to include novel goals, such as climate change adaptation and mitigation (Aitken et al., 2008). Understanding diversity drivers associated with management will aid in the development of flexible, varied strategies and tools for adapting management to future conditions (Millar et al., 2007).

Most studies examining forest composition/structure and driving factors either characterize in greater detail a relatively small number of stands over a limited geographic extent (e.g., Hupperts et al., 2020; Kern et al., 2013; Sabo et al., 2019; Walters et al., 2016) or cover a broad geographic and temporal extent but are limited in plot size and therefore applicability at the stand level (e.g., studies utilizing U.S. Forest Service Forest Inventory

and Analysis data; Baach et al., 2022; Bose et al., 2016; Tinkham et al., 2018). Few studies have analyzed stand-level tree composition/structure data at regional (10^6 ha) scales, which can capture the effects of potentially important driving factors that vary at this scale (e.g., site quality, deer density). In this study, we used a forest structure dataset from 141 selection-managed NHFs, distributed throughout northern Michigan, USA, to address knowledge gaps in understanding the patterns and potential causes of current and potential future diversity and composition of NHFs. Our objectives are to quantify patterns of diversity and absolute and relative density of key individual species by and among size classes (seedling–sapling–canopy); to characterize associations of these patterns with possible stand and landscape drivers; and to interpret these patterns and relationships in terms of possible future trajectories for NHF composition and diversity. Based on the current knowledge of general patterns and expected effects of multiple factors driving regeneration (seedling and sapling classes), in managed NHFs, we predict the following:

1. Tree diversity will be highest on sites with (a) lower site quality and (b) greater canopy tree diversity. Among size classes, diversity will be greatest in (c) the canopy class, followed by seedlings and lastly saplings.
2. For individual species, factors limiting establishment, survival, and recruitment of tree regeneration will differentially affect the density of key regeneration size classes, thereby creating signature patterns of relative abundance among size classes (e.g., a sapling recruitment bottleneck leads to lower sapling relative abundance compared with seedling and canopy). We predict regeneration will be governed as follows by key drivers:
 - (2a) Conspecific canopy stem density (a proxy for local seed source density) will associate positively with density of regeneration, particularly for seedlings of large-seeded, passively dispersed species. Thus, if our study landscape is characterized by low stand-scale canopy diversity, regeneration diversity will be low too.
 - (2b) Small-seeded species seedling abundance will associate positively with coverage of coarse woody debris (CWD) and BMS substrates and negatively with hardwood litter (HWL) coverage. Across our sampled landscape, we predict past management has led to HWL dominance of the forest floor, leading to a general pattern of low seedling and sapling abundance relative to canopy trees for small-seeded species.
 - (2c) Stand density (basal area [BA], a proxy for light availability) will negatively associate with
 - sapling density, particularly for saplings of shade-intolerant species. We predict low sapling density relative to canopy and seedling classes across the landscape for shade-intolerant species, given current management practices including the high BA of the ready-to-harvest stands we sampled.
 - (2d) Greater stand-scale deer use will negatively associate with sapling densities of palatable tree species. Thus, we expect lower relative densities of saplings than seedlings and canopy trees for palatable species to be the general landscape pattern, given sustained high deer use over large extents of the landscape we sampled.

METHODS

Study area and data collection

Our data are from 141 managed NHF stands distributed throughout northern Michigan, which are part of a larger experiment evaluating silvicultural alternatives to STS management (Walters, Roloff, et al., 2020). Here, we focus on pre-harvest forest structure. Most stands we sampled established as even-aged stands following exploitive intense harvests in the early 20th century (Henry, 2022). At the time of sampling, the stands were considered ready or near ready for a commercial partial harvest by STS criteria (i.e., $>23 \text{ m}^2 \text{ ha}^{-1}$ BA and well-stocked in sawtimber classes; Arbogast, 1957) and were likely subject to 3–4 partial/selection harvests (beginning in 1960s) prior to our sampling them (Bernie Hubbard MDNR retired, personal communication). Additional vegetation sampling details can be found in the study by Henry et al. (2021); we recount pertinent details here.

Most stands were state owned ($n = 119$) and managed by the Michigan Department of Natural Resources (MDNR), with the remaining 22 stands owned and managed by private forest industry companies (Hancock Timber Resource Group, The Rohatyn Group). Stands were generally present on upland mesic to wet-mesic sites, with fertile medium-textured soils (Dickmann & Leefers, 2016) and dominated by sugar maple (vast majority), red maple, or basswood (see Table 1 for scientific names and authority). Hereafter we refer to species by their common name or acronym. We established a systematic grid of 25 survey points within a 12-ha square within each stand and collected species-level data in the following plots at each survey point. We counted all tree seedlings shorter than 50 cm and ocularly estimated the percent cover of shrubs and groundcover composition (e.g., HWL, BMS, downed woody material) within two quadrats, each 1 m^2 (50 m^2 total per stand). Within a

TABLE 1 Summary statistics (mean and maximum) for stand-level tree density (in stems per hectare) by species and size class (plus total occurrences, *n*, across all size classes) for species with more than 14 stand-level occurrences (*n*).

Species	Seedlings			Saplings			Canopy			All <i>n</i>
	Mean	Max	<i>n</i>	Mean	Max	<i>n</i>	Mean	Max	<i>n</i>	
<i>Abies balsamea</i> [L.] Mill.* (balsam fir, BF)	817	11,295	47	154	936	64	41	279	62	77
<i>Acer rubrum</i> L.* (red maple, RM)	17,823	139,733	118	641	5673	71	86	460	84	119
<i>Acer saccharum</i> Marsh.* (sugar maple, SM)	71,797	448,752	140	1217	6943	140	282	531	139	141
<i>Betula alleghaniensis</i> Britton* (yellow birch, YB)	609	3664	47	81	725	39	15	113	66	85
<i>Betula papyrifera</i> Marsh. (paper birch, PB)	350	1055	9	10	32	7	10	28	20	27
<i>Fagus grandifolia</i> Ehrh.* (American beech, AB)	4090	49,171	89	1671	5903	96	41	195	87	98
<i>Fraxinus americana</i> L.* (white ash, WA)	7885	55,453	71	451	5188	43	12	67	31	72
<i>Ostrya virginiana</i> Mill.* (ironwood, IW)	2744	14,566	108	955	4545	111	17	74	77	116
<i>Picea</i> spp. (spruce, SP)	244	2159	22	25	223	35	9	39	32	53
<i>Pinus strobus</i> L. (white pine, WP)	279	2059	22	77	700	17	9	21	14	30
<i>Populus grandidentata</i> Michx. (big tooth aspen, BTA)	459	1495	5	73	127	7	12	67	13	20
<i>Populus tremuloides</i> Michx. (quaking aspen, QA)	342	833	9	98	421	11	11	57	23	31
<i>Prunus pennsylvanica</i> L.f. (pin cherry, PC)	497	1632	6	70	396	10	6	11	3	16
<i>Prunus</i> spp.* (cherry, CH)	3283	31,632	112	126	863	73	32	191	59	118
<i>Quercus rubra</i> L.* (red oak, RO)	1166	3382	35	138	605	13	16	64	26	37
<i>Tilia americana</i> L.* (basswood, BW)	884	3832	65	89	732	31	49	279	78	87
<i>Tsuga canadensis</i> [L.] Carr. (eastern hemlock, EH)	253	864	13	44	138	13	16	88	29	35
<i>Ulmus americana</i> L.* (American elm, AE)	441	2400	31	47	389	20	7	21	20	47

Note: Means were calculated from occurrence data only: for example, subset of stands where a species size class combination was present. Species labeled with * were modeled for individual density by size class. A full version of this table including species with less than 14 occurrences can be found in Appendix S1: Table S2.

2-m-radius circular plot (314 m² total per stand), we tallied stems 50–137 cm tall and tallied and measured dbh for stems >137 cm tall and <5 cm dbh. Within a 6-m-radius circular plot (2827 m² total per stand), we tallied and measured dbh for all stems >5 cm dbh. From these data, we established the following size classes: seedlings, 0–137 cm tall; saplings, >137 cm tall and <10 cm dbh; and canopy trees, ≥10 cm dbh. The rationale behind establishing seedling strata was to reflect seedling establishment limitations (local seed availability, germination/early establishment substrate availability); the sapling strata, which represent possible canopy recruits and are additionally impacted by seedling to sapling recruitment limitations (i.e., deer browsing, shade-driven limitations to growth and survival; Walters, Farinosi, & Willis, 2020); and the canopy tree class, which represents the dominant stratum of mature/adult trees and thus seed sources for regeneration. Notably, some subcanopy species mature at smaller size and have relatively small maximum size (e.g., ironwood, serviceberry). For all size classes, we calculated density as stem counts per hectare. In addition, for canopy trees, we calculated density as the cross-sectional area of boles at breast height summed

across stems (BA, in square meters per hectare). Given our focus on stand-level metrics, we aggregated plot data within each stand and analyzed data at the stand level.

At the stand scale, we assigned a dominant habitat type, an index of site quality, based on the assessment of herbaceous plant assemblages and key indicator species (Burger & Kotar, 2003). We assessed winter deer use at the stand scale with fecal pellet transect surveys in the spring of 2017 at a subset of 50 sites, surveying 628 m of transect (6 m wide) subdivided into 103 segments at each site. We modeled estimated winter deer use (percent transect segments occupied by deer pellets) for all stands in 2017 with a spatial model in a Bayesian framework; the model also incorporated well-established climate and landscape factors known to influence local deer populations, plus deer pellet transect surveys conducted in 2019 at 139 sites following experimental harvest (Henry et al., 2021).

Species diversity analyses

All analyses and modeling were conducted in R (R Core Team, 2018). To characterize species diversity, we used

Hill numbers, a unified set of diversity measures parameterized by a diversity order q , which correspond to common diversity metrics ($q = 0$ is directly equivalent to species richness [SR]; $q = 1$ is equal to the exponential of the Shannon diversity index; $q = 3$ is equal to the inverse of Simpson diversity). Hill numbers have several advantages, including obeying doubling properties (e.g., when two communities with the same number of equally abundant but distinct species are combined, the measured diversity value is double that of the original communities) and being interpretable in units of effective number of species, which can be thought of as the number of species that would need to be equally abundant to generate the same level of diversity (Chao et al., 2014). For example, if you had a community with a total of 15 species ($q = 0$) and a Shannon diversity index of 1.95, in a community with equally abundant species, there would need to be 7 species (Hill number $q = 1$) present to generate the same level of diversity. Hill numbers also make differences between communities more apparent; for example, given the same number of individuals, if most of the 15 species ($q = 0$) were equally abundant, Hill numbers would decline more gradually with increasing q , for example, Hill $q = 1$ of 13 effective common species (CS), and comparing 7 and 13 is more intuitive than 1.95 and 2.56 (Shannon diversity index of second community). Given low SR of our stands, we focus on $q = 0$ (SR) and $q = 1$ (effective number of CS) (Hsieh et al., 2016).

We were limited in assessing stand-scale diversity by having sampled a fixed area, much smaller than the stands in question, for all size classes. Although we sought to achieve similar sampling intensity across size classed by using larger sample plots for larger size classes, imbalances in sampling intensity across size classes remained. Due to these sampling limitations, we used the iNEXT package (Hsieh et al., 2016) to conduct rarefaction and extrapolation with our data to estimate asymptotic Hill numbers (Chao et al., 2014). Rarefaction accounts for sampling completeness and involves generating a rarefaction curve, which shows the number of species as a function of sampling intensity (number of individuals or number of sampling units). iNEXT conducts rarefaction and then extrapolates the curve to estimate the value at the curve asymptote, representing the theoretical SR (i.e., achieved in a complete census rather than a sample). We analyzed data as abundance values (total number of individuals sampled per species) rather than incidence values (total number of plot occurrences per species), given the relatively small number of plots (25). We used the iNEXT function to calculate SR and CS for all three size classes.

We conducted generalized linear models of diversity metrics, by size class, as a function of landscape and

site-level factors that we standardized (mean centered and divided by the SD). These factors and their rationales were as follows: winter deer use (DEER) as a proxy for browsing pressure, average stand-level BA (in square meters per hectare of trees ≥ 10 cm dbh) as a proxy for shading/decreasing light, standard deviation of stand-level basal area (BA_{SD}) as a proxy for heterogeneity in light availability, asymptotic species richness (SR10) and effective common species (CS10) of the canopy stratum as proxies for species heterogeneity in local seed availability, three forest floor cover classes representing potential substrates for/impediments to seedling establishment (percent cover of HWL, BMS, and CWD), and shrub coverage (shrub) to represent shrub competition intensity with seedlings. There were extreme outlier values for HWL (2) and BMS (1), which were more than 6 SDs from the mean; we truncated them to the closest nonoutlier value as they represent unusual stand conditions that we did not sample well and could potentially have high leverage in our models. Substrate and shrub cover were only used for seedling class models (Table 2). While we focus on the three most important groundcover classes (HWL, BMS, and CWD), there were a variety of other groundcover classes with minor coverages recorded such that BMS, CWD, and HWL do not sum to 1. We calculated variance inflation factors (VIF), a measure of multicollinearity among predictors in linear regression models (Marquardt, 1970), using `check_collinearity` in the performance package (Lüdtke et al., 2020); at a threshold of 5, below which predictors are generally considered to have low collinearity (James et al., 2013), none warranted removal.

In a Bayesian framework, we used a gamma distribution with a log link to model estimated asymptotic SR and CS (both of which are continuous positive). We ran models using the rjags package (Plummer, 2018), which interfaces JAGS (Just Another Gibbs Sampler) software (Plummer, 2003) with R. We ran models for 20,000 iterations after a 5000-step adaptation period with three chains, which, based on our visualizing trace plots and calculating Gelman's Diagonal, was sufficient for model convergence. We used the coda package for model convergence diagnostics (Plummer et al., 2006).

We evaluated associations with site quality through a separate analysis to preserve df. To determine whether ordinal site quality diversity measures were comparable between the three broad geographic regions of our study, we compared generalized linear models of diversity predicted by site quality versus a site quality–region interaction, using gamma distributions. We compared the simple model versus the interaction model using the Akaike information criterion (AIC), which is an estimator of prediction error (Akaike, 1974). The interaction model had a

TABLE 2 Summary statistics for stand-level predictor variables included in seedling and sapling models of diversity and individual species density.

Predictor	Acronym	Mean	Min	Max
Deer use (% segments occupied)	DEER	19	2	60
Stand basal area, canopy trees ($\text{m}^2 \text{ha}^{-1}$)	BA	26	20	44
Standard deviation of stand basal area ($\text{m}^2 \text{ha}^{-1}$)	BA_{SD}	14	7	23
Density of conspecific canopy trees (stems ha^{-1})	SeedS	a	a	a
Species richness of canopy trees	SR10	7	1	20
Effective common species of canopy trees	CS10	3	1	7
Shrubs (% cover) ^b	SHRUB	2	0	26
Bare mineral soil (% cover) ^b	BMS	1	0	8
Hardwood litter (% cover) ^b	HWL	92	72	97
Coarse woody debris (% cover) ^b	CWD	3	1	7

Note: A short explanation of variables includes the following: DEER, estimated deer browsing pressure; BA, an index of shading by dominant canopy; BA_{SD} , an index of light/shading heterogeneity; SeedS, an index of mature, canopy tree seed source density; SR10 and CR10 are estimates of species heterogeneity in seed sources; SHRUB is an index of shrub competition for seedlings; and BMS, HWL, and CWD are indices of availability/coverage of seedling establishment facilitating/impeding substrates. Summaries for BMS and HWL were calculated after removing outlier values.

^aSee Table 1 (canopy).

^bSHRUB, BMS, HWL, and CWD were only used as predictors for the seedling models.

lower AIC for all three size classes and both diversity metrics. Since we were primarily interested in the effect of site quality, we used the emmeans function to estimate marginal means and the pairs function to conduct pairwise comparisons across site quality for each region separately (Lenth, 2020).

Generalized linear modeling for species density

We included 11 species present in >10% of stands (>14 stands) for each of the three size classes. We included the same predictors as in the species diversity models (Table 2), which we again standardized (mean centered and divided by the SD), plus an additional predictor to measure the density of conspecific stems ≥ 10 cm dbh (SeedS), a proxy for seed source (McEuen & Curran, 2004). We did not include site quality as a predictor because our chosen index of site quality and tree composition are confounded and because tree composition is included in the model in the SeedS predictor (Burger & Kotar, 2003). We compared two candidate models: a negative binomial generalized linear model and a zero-inflated negative binomial generalized linear model. We checked for multicollinearity, simulated models in a Bayesian framework, and checked for model convergence using the same procedures as described for species diversity metric models. To select a model, we compared deviance information criteria (DIC) values, which incorporate model fit and complexity (Spiegelhalter

et al., 2002). DIC comparison led to model selection of a negative binomial model for seedlings and saplings of AB, CH, IW, RM, and SM, plus BW seedlings. BW saplings plus AE, BF, RO, WA, and YB were modeled with a zero-inflated negative binomial model following DIC comparison (Appendix S1: Table S1). We reported selected model results at $p < 0.10$ given relatively low number of stands, coarse aggregation of data, and our intention to highlight trends.

Relative density as a function of size class

For each stand, we calculated the relative density for each present species by size class (stems of a given species divided by total stems). We limited our relative density by size class analyses to the 18 tree species present, in any size class, on at least 10% of stands (>14 stands). In this case, relative density is useful because all three size classes are portrayed on the same scale, independent of stem density, which varies widely between seedlings and canopy trees; it also has the benefit of reflecting potential shifts in composition among classes as it places species-specific density in the context of competitors. For each stand, we extracted the rank order of species density, by size class, for each species present. For example, in a hypothetical stand, sugar maple has the highest relative density in canopy/mature (referred to as canopy for brevity) stems (0.7), followed by seedlings (0.6) and saplings (0.4), making the ranking “canopy–seedling–sapling”; in the same stand, hemlock had a relative

density of 0.01 in the canopy, and 0 for seedlings and saplings, making the ranking just “canopy.” We then tallied the frequency of each relative density ranking among stands, for each species. For example, sugar maple had the ranking “canopy–seedling–sapling” in X stands, “seedling–canopy–sapling” in Y stands, and so on. We only included sites in which a species was present in at least one size class for this analysis. This resulted in $n < 141$ for all species except sugar maple, which was present on all sites. We used a Kruskal–Wallis rank sum test to determine whether relative density varied by size class, and, if significant, pairwise Wilcoxon tests for paired (within stand) values with a Bonferroni correction ($n = 48$) to determine individual differences. To aid in generalizing differences and similarities in species patterns of relative density among size classes, we labeled seedling–sapling–canopy distributions with four shape categories: left-skewed, right-skewed, positive parabola, and negative parabola.

RESULTS

Overview of stand characteristics

Our stands indicated typical characteristics of mature, ready-to-harvest managed NHFs: relatively high BA dominated by sugar maple, with low shrub coverage and high proportions of HWL substrate. BA of trees ≥ 10 cm dbh averaged $28 \text{ m}^2 \text{ ha}^{-1}$, and BA_{SD} averaged $14 \text{ m}^2 \text{ ha}^{-1}$ (Table 2). Mean cover of SHRUB was 2%, and HWL cover (mean 91%) dominated forest floor substrate categories (BMS, and CWD each $< 3\%$). DEER was variable, ranging from 2% to 60% pellet occupancy of transect segments. We documented 25 tree species in our survey (Appendix S1: Table S2). Total stand-level stem density (in stems per hectare) ranged from 9605 to 460,112 (mean 99,033) for seedlings, 357 to 12,771 (mean 3797) for saplings, and 219 to 778 (mean 449) for canopy/mature stems (≥ 10 cm dbh). At least nine species were sampled in any size class on at least half of the sites, including sugar maple, which was sampled in all stands (Table 1). Summing across all 141 stands, nine species had an average relative density of at least 1% in at least one size class (Figure 1). Although sugar maple had the highest average relative density for seedlings, saplings, and canopy/mature trees, the relative density of ironwood and American beech was nearly equivalent to that of sugar maple for the sapling class (Figure 1).

There was considerable variation in estimated diversity metrics. Among stands and size classes, the highest estimated asymptotic SR was 20 and the lowest was 1, both occurring in the canopy class (Figure 2). Among size classes, average SR was highest for seedlings (7.6), followed by canopy/mature stems (7.4) and saplings (6.4),

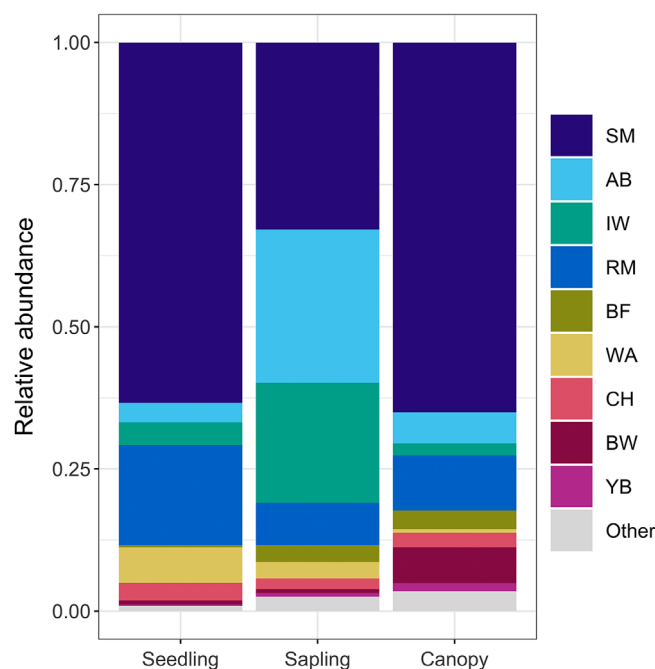


FIGURE 1 Landscape-level (i.e., data from 141 stands in Michigan) average relative density, by size class. Species not achieving at least 1% relative density in any size class were grouped as “Other.” Species codes and list of common species are in Table 1, and the full list of species is in Appendix S1: Table S2. Seedlings are 0–137 cm tall, saplings are > 137 cm tall and < 10 cm dbh, and canopy/mature are ≥ 10 cm dbh.

though only seedlings and saplings significantly differed from one another ($p < 0.05$) (Figure 2). Effective (i.e., common) species values averaged 3.2 for saplings (highest), 2.8 for canopy/mature trees, and 2.4 for seedlings (lowest), with all size categories different from one another ($p < 0.05$) (Figure 2); despite saplings having the lowest SR, they had the highest evenness.

Species diversity measures predicted by stand and landscape factors

The diversity of tree regeneration was generally related to canopy composition and structure but generally unrelated to the cover of different forest floor substrates, competing shrub cover, deer use, or site quality (Table 3). Seedling SR and CS were generally positively associated with canopy diversity (CS10), and sapling SR was negatively associated with BA (though only significant at the 90% credible interval for SR). For forest floor substrate variables, seedling SR declined with BMS cover and otherwise had no significant associations. Diversity indices typically did not vary with site quality, except for the effect of CS declining for seedling and canopy strata in the eastern Upper Peninsula (Figure 3).

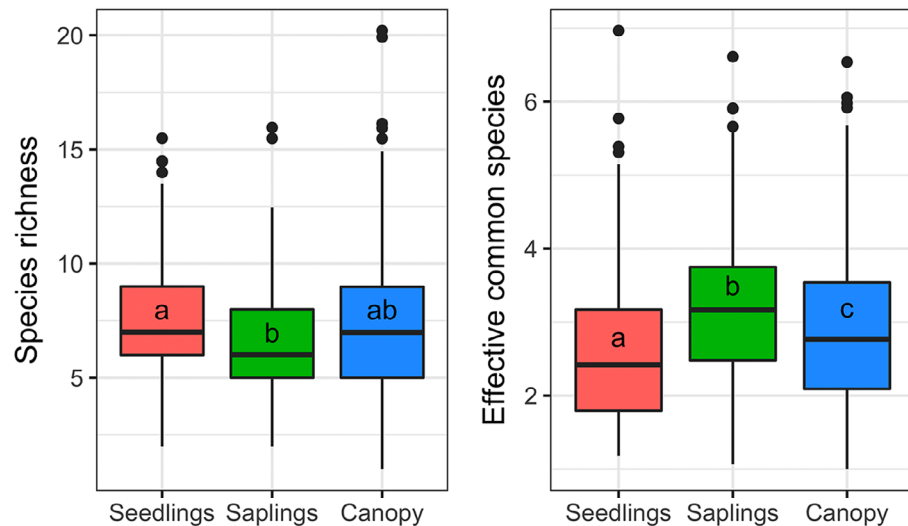


FIGURE 2 Stand-level estimated asymptotic species richness and effective number of common species for seedlings (0–137 cm tall), saplings (>137 cm tall and <10 cm dbh), and canopy/mature trees. Box plot hinges show 25th percentile, median, and 75th percentile. Whiskers extend up to 1.5 times the interquartile range; values beyond that are shown as outlier dots. Within each panel, box plots labeled with the same letter are not significantly different. Note different y-axis limits between the two plots.

TABLE 3 Parameter estimates (median and a 95% credible interval, analogous to a slope term) for associations between model predictors and species richness/common species for seedlings (0–137 cm tall) and saplings (>137 cm tall and <10 cm dbh).

Predictor	Species richness		Common species	
	Median	95% CI	Median	95% CI
Seedlings				
Intercept	2.022	(1.969 to 2.075)	0.967	(0.904 to 1.032)
Shrub	0.035	(−0.024 to 0.1)	−0.025	(−0.097 to 0.051)
DEER	0.018	(−0.037 to 0.075)	0.001	(−0.066 to 0.071)
BA	0.005	(−0.056 to 0.066)	−0.003	(−0.072 to 0.066)
BA _{SD}	0.037	(−0.023 to 0.098)	0.022	(−0.045 to 0.088)
BMS	−0.07	(−0.137 to −0.004)	−0.003	(−0.084 to 0.079)
HWL	−0.037	(−0.119 to 0.042)	−0.036	(−0.127 to 0.052)
CWD	−0.006	(−0.068 to 0.057)	−0.021	(−0.097 to 0.055)
CS10	0.047	(−0.018 to 0.111)	0.102	(0.022 to 0.184)
SR10	0.096	(0.033 to 0.162)	−0.009	(−0.086 to 0.071)
Shape	9.852	(7.697 to 12.419)	6.809	(5.32 to 8.561)
Saplings				
Intercept	1.843	(1.785 to 1.903)	1.156	(1.104 to 1.209)
DEER	−0.038	(−0.097 to 0.021)	−0.005	(−0.058 to 0.049)
BA	−0.06	(−0.124 to 0.005)	−0.032	(−0.089 to 0.026)
BA _{SD}	0.009	(−0.055 to 0.073)	0.003	(−0.054 to 0.06)
CS10	0.137	(0.066 to 0.208)	0.122	(0.06 to 0.184)
SR10	0.049	(−0.02 to 0.121)	0.03	(−0.03 to 0.092)
Shape	7.983	(6.259 to 10.02)	10.282	(8.051 to 12.877)

Note: Predictor acronyms are defined in Table 2, except for shape, which defines the gamma distribution. Significant associations (95% CI) are bolded; associations significant only at a 90% CI are italicized.

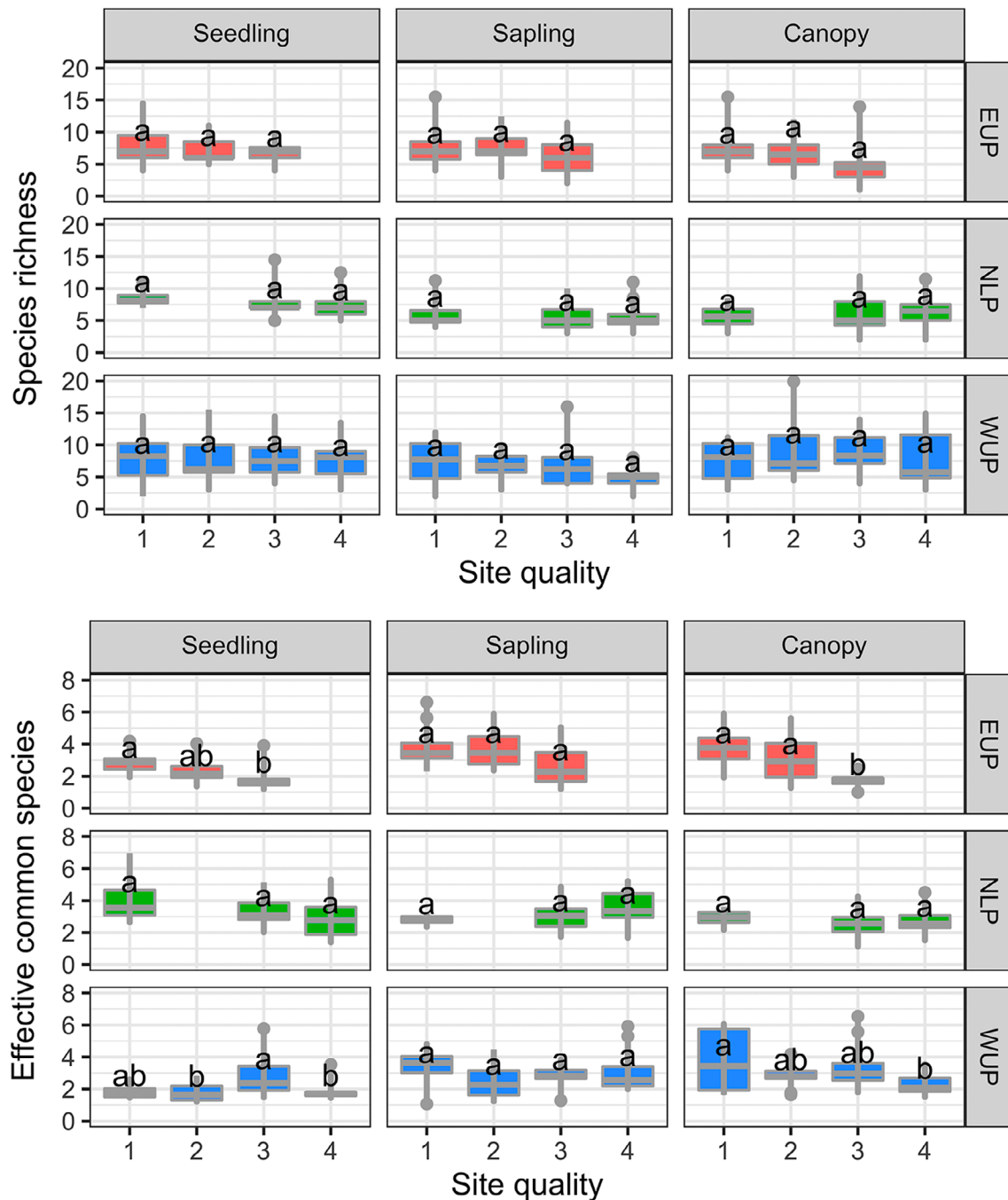


FIGURE 3 Box plots of asymptotic species richness and number of common species by site quality and region. For site quality, 1 = lowest quality and 4 = highest (Burger & Kotar, 2003). Box plot hinges show 25th percentile, median, and 75th percentile. Whiskers extend up to 1.5 times the interquartile range; values beyond that are shown as outlier dots. Within panels, bars with the same letters are not significantly different based on pairwise comparison of estimated marginal means. Note different y-axis limits between the two plots.

Individual tree species density predicted by stand and landscape factors

Similar to models of diversity metrics, the density of NHF seedlings and saplings by species were also most consistently associated with canopy structure, though additional significant associations were detected with deer

browsing pressure, shrub coverage, and substrate (Figure 4). Balsam fir seedling densities were positively associated with BA, while saplings of red maple, sugar maple, and cherry species were negatively associated with BA. Maple species had higher densities in stands with greater BA_{SD} , while balsam fir had negative associations. SeedS had, by far, the most consistent associations

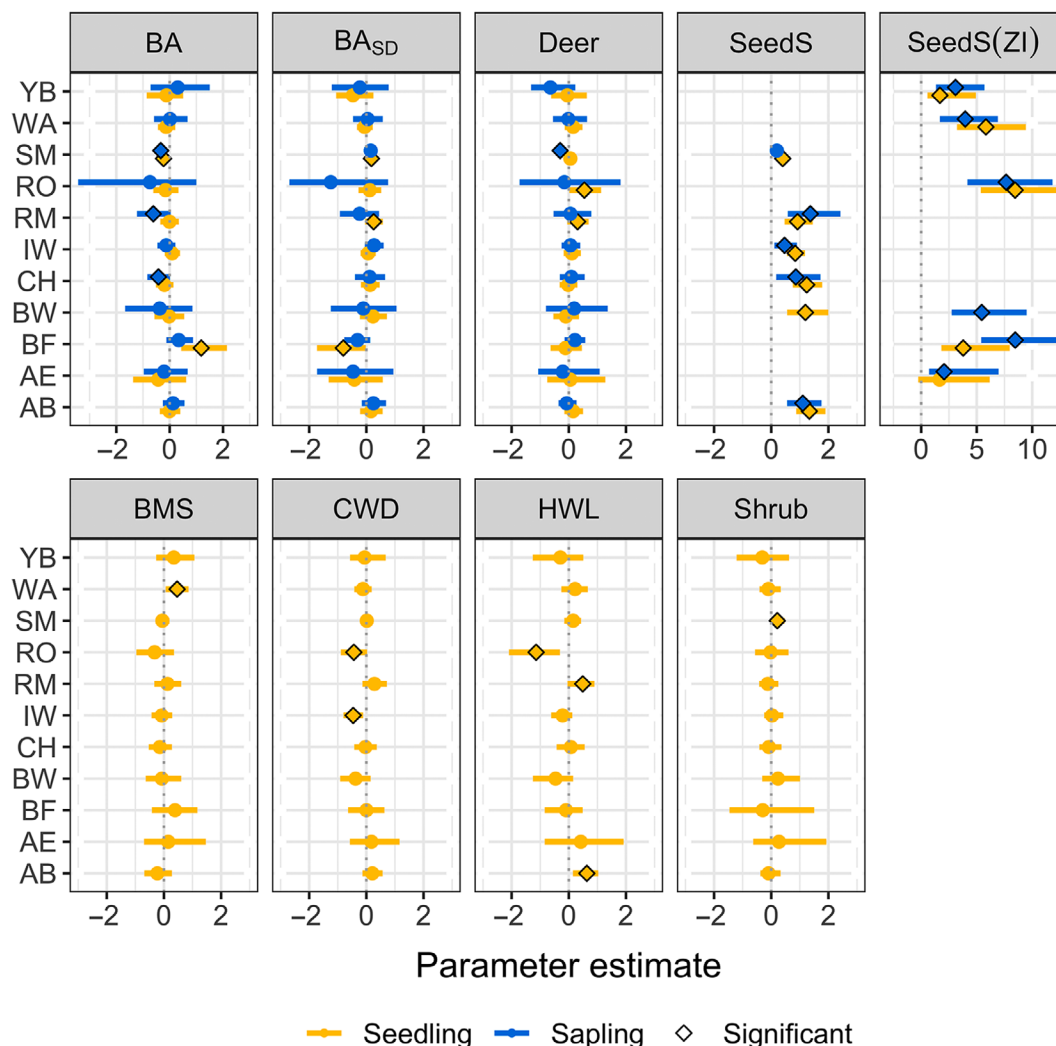


FIGURE 4 Parameter estimates (median with 0.05–0.95 credible interval) for predictor variables (acronyms defined in Table 2) with seedling (0–137 cm tall) and sapling (>137 cm tall and <10 cm dbh) densities by tree species (codes defined in Table 1). Substrate (BMS, CWD, HWL) and shrub predictors were only tested for seedlings. Median values of significant ($p > 0.10$) parameters are diamonds, outlined in black. Negative binomial models were used for seedlings and saplings of AB, CH, IW, RM, and SM and for seedlings of BW, and zero-inflated negative binomial models were used for the others. Model parameters were standardized so scales are comparable; however, SeedS for the zero-inflated models (SeedS(ZI)) were included as presence/absence predictors and are not directly comparable to other parameters, so note the different x-axis limits for SeedS(ZI).

among factors potentially driving species-specific densities, with positive associations for all species in both seedling and sapling classes. DEER associations were overall weak, negative only for sugar maple saplings, and positive for red maple and red oak seedlings. Of factors related only to seedling densities, SHRUB effects were only significant for sugar maple (+) and significant substrate effects were few and lacked generalization.

Species relative density across size classes

Among species, size class was a significant predictor of relative abundance for 16 of 18 species examined, with 13

species having at least one significant pairwise comparison, and patterns were generally consistent with species life history strategies and functional traits (Figure 5). Size class distributions were left-skewed (relative density increasing as a function of size) for seven species (e.g., eastern hemlock, yellow birch), a negative parabola shape (saplings having highest relative density) for six species (e.g., American beech, ironwood), positive parabola (saplings least abundant class) for two species (sugar maple and red oak), and right-skewed (relative density declining with size class) for three species (white ash, red maple, and cherry species; Figure 5).

Our relative abundance results are further supported by summaries of relative density rankings among size

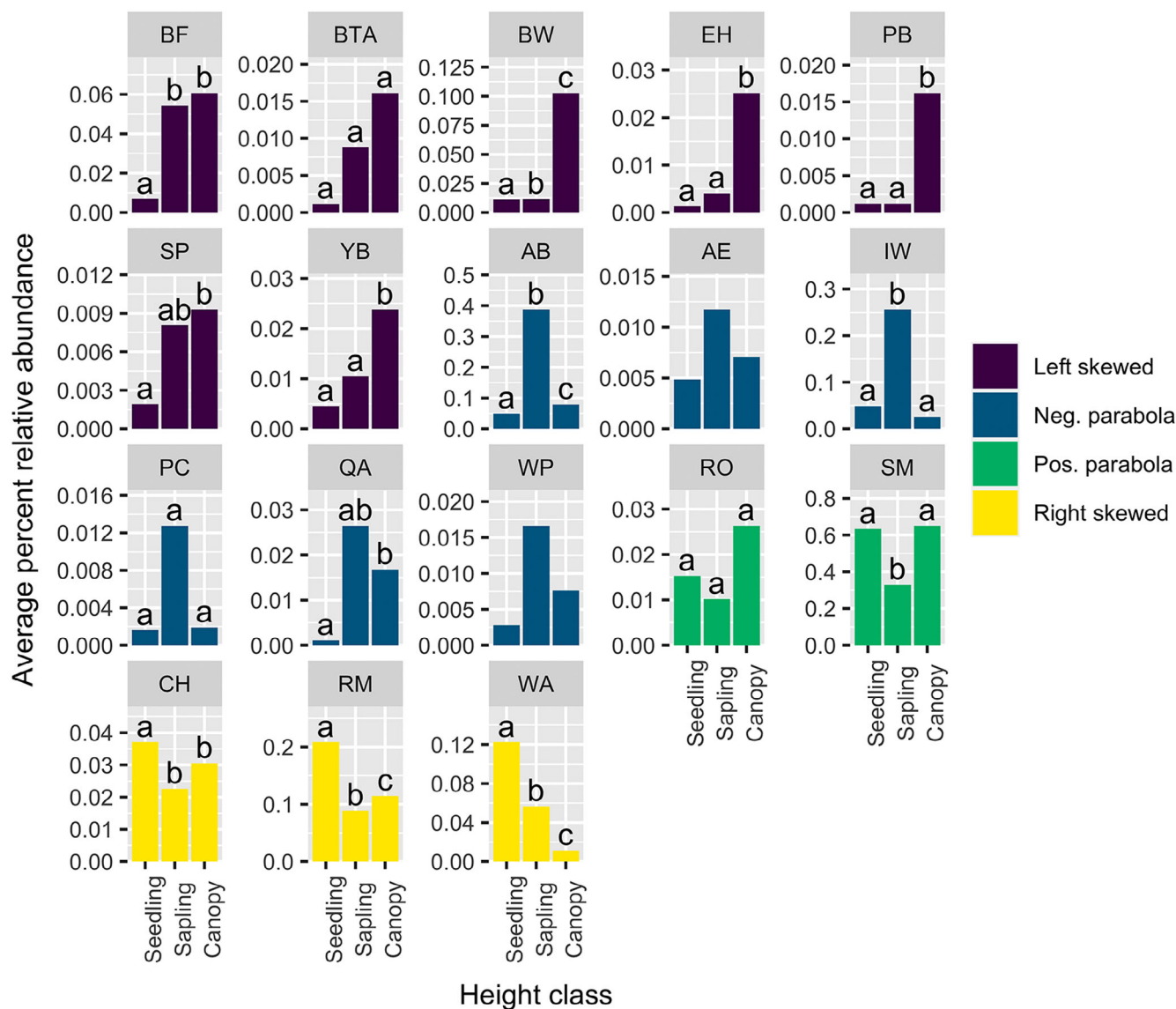


FIGURE 5 Relative density by species (stems relative to total stems, all species) for each size class for seedlings (0–137 cm tall), saplings (>137 tall and <10 cm dbh), and canopy trees (≥ 10 cm dbh) strata; for each species, we only included the stands where stems were present in at least one size class. Species codes are defined in Table 1. Within a species plot, different letters indicate significant differences in relative density between size classes (pairwise Wilcoxon tests, $p < 0.05$). Size class was not a significant predictor for American elm and white pine, so we did not calculate pairwise comparisons. Color corresponds to their identified shape across size classes (key on right). y axes vary by plot.

classes for each stand, which confirm that the overall observed patterns are repeatedly observed in individual stands rather than being driven by outlier stands skewing the average (Table 4). Stand-by-stand variation reveals additional details in support of the landscape patterns (Figure 5). For example, left-skewed species are defined by canopy trees as the dominant relative density category (Figure 5), and this pattern is generally supported by the frequency of relative density patterns at the level of individual stands (for first and second most common patterns, canopy trees ranked first, Table 4). However, for individual stands, it is striking that for four of the seven species

(yellow birch, paper birch, big tooth aspen, and eastern hemlock) the most common stand-/site-level relative density pattern was to have only canopy trees in the sample (i.e., no saplings or seedlings). Similar but oppositely for right-skewed species for two of the three species (white ash and cherry species), the most common relative density pattern at the stand level was for samples to contain only seedlings. Although lack of sampled stems can be attributable to very low density, leading to lack of detection in our samples (rather than actual true absence), these patterns reinforce how stark the differences in relative abundance can be among size classes within a given stand.

TABLE 4 Summaries of stand-specific relative density rankings by size class for 18 species surveyed in >10% of our 141 stands.

Shape	Species	<i>n</i>	Seedling %	Sapling %	Canopy %	First ranking		Second ranking	
						Order	<i>n</i>	Order	<i>n</i>
Left-skewed	BF	77	0.09	0.32	0.58	Can-Sap-Seed	21	Can-Sap	15
	YB	85	0.16	0.14	0.69	Can	23	Can-Seed	14
	PB	27	0.19	0.07	0.74	Can	13	Seed	5
	SP	53	0.11	0.4	0.49	Sap	14	Can	9
	BTA	20	0.15	0.3	0.55	Can	8	Sap	4
	BW	87	0.11	0.05	0.84	Can-Seed	29	Can	17
	EH	35	0.09	0.09	0.83	Can	15	Can-Sap-Seed	6
Negative parabola	AB	98	0.01	0.95	0.04	Sap-Can-Seed	60	Sap-Seed-Can	20
	IW	116	0.05	0.86	0.09	Sap-Seed-Can	33	Sap-Seed	31
	WP	30	0.3	0.37	0.33	Seed	7	Sap-Seed	5
	QA	31	0.06	0.29	0.65	Can	14	Sap	5
	PC	16	0.31	0.5	0.19	Sap	7	Seed	5
	AE	47	0.51	0.23	0.26	Seed	17	Can	6
Positive parabola	SM	141	0.46	0.11	0.43	Seed-Can-Sap	51	Can-Seed-Sap	43
	RO	37	0.41	0.14	0.46	Seed	10	Can-Seed	9
Right-skewed	RM	119	0.8	0.08	0.13	Seed-Can-Sap	35	Seed	31
	WA	72	0.78	0.18	0.04	Seed	19	Seed-Sap	15
	CH	118	0.55	0.25	0.2	Seed	31	Sap-Seed	17

Note: Species are grouped by the shape of their average relative density size class distribution, and *n* is the number of sites that a species was present on, in any size class. Seedling, sapling, and canopy % indicate the percentage of stands in which that size class had the highest relative density. First and second rankings refer to the two most common rankings of relative densities by size class, in decreasing rank order (i.e., the first size category listed had the highest relative density), with empty size classes omitted.

Stand summaries also indicate variation in typical stand-level patterns of species within the same group. For example, where they locally occur across the landscape, red oak and sugar maple both have lower relative density of saplings than seedlings and canopy trees (positive parabola), but saplings are still present (relative density = 0.32 for SM and 0.01 for RO). However, while sugar maple is most commonly present in all size classes (relative density of seedling > canopy > saplings), considerably less dense red oak most often had only seedlings and secondarily, canopy stems and seedlings. While sugar maple typically has a seed source present, red oak appears to sometimes rely on long-distance seed dispersal while also having a sapling recruitment bottleneck.

DISCUSSION

Diversity patterns and drivers

Despite sampling 25 tree species across the NHF landscape, most species had limited stand-level occurrences and low density; fewer than 10 species were sampled in a

majority of stands or comprised >1% of stems in the overall data set. Our diversity metrics are similar or slightly lower than past studies of NHFs. However, contextualizing our results in relation to past literature is hampered by inconsistencies with regional study scale, sampling intensity, and approach for diversity metric calculations. For example, average stand-level tree SR in this study, for all size classes, was lower than values reported for managed and old growth stands in Crow et al. (2002). However, Crow et al. covered a smaller geographic range than analyzed here and reported values for aggregates of three stands, which may not be comparable to single-stand estimates if species turnover among stands (beta diversity) was high. Our estimates were lower than Danyagri et al. (2019) report, although this comparison relies on their modeled estimates of range and mean versus our actual estimates of mean and range which account for sampling intensity. Standardization of diversity metric calculations, such as the use of Hill numbers (Chao et al., 2014), would benefit future comparisons of diversity metrics across studies.

Comparison of results to other studies also highlights spatial heterogeneity of tree regeneration in northern

hardwoods. We showed similar dominance of the seedling class but lesser dominance of the sapling class by sugar maple compared with Neuendorff et al. (2007) and Crow et al. (2002), studies which were conducted in a limited geographic region in the western Upper Peninsula of Michigan known to support copious sugar maple regeneration (Henry et al., 2021). Rather, our sapling compositions, from a much broader region, are characterized by roughly even representation of sugar maple, ironwood, and American beech (Figure 1). This pattern is similar to that found by Angers et al. (2005), near Ottawa, Canada, and Elenitsky et al. (2020), in northeastern Michigan, both of which are in regions of relatively lower lake-effect snowfall and therefore likely higher deer browsing pressure (Henry et al., 2021; Shi et al., 2006). Our results highlight the importance of considering patterns over broad geographic scales to inform landscape-level management recommendations.

Regarding our first prediction, tree regeneration diversity was similar across the range of site qualities we surveyed (failing to support our prediction 1a). Patterns of diversity showed trends of modest decreases in diversity with increasing site quality in some cases, but only significantly for seedling and canopy classes in the eastern Upper Peninsula region. Lack of significant associations could be due to the low average richness values in our stands, as differences among stands are likely to be small and may require greater statistical power than we were able to achieve. It also may be that, while species compositional shifts among site qualities are well documented in our region (e.g., more sugar maple on higher quality sites, more red maple on lower quality sites; Burger & Kotar, 2003; Elenitsky et al., 2020), changes to diversity are negligible to modest on the range of site qualities we examined. However, regeneration diversity was higher in stands with greater canopy diversity (Table 3), offering clear support for prediction 1b. Our findings support previous results indicating canopy composition and diversity are key drivers of tree regeneration diversity (Bose et al., 2016; Tinya et al., 2019). Canopy-effective CS were generally stronger predictors of regeneration diversity (except for seedling SR, better predicted by canopy SR), suggesting that more even relative densities among canopy species rather than presence alone are key for driving diverse regeneration. An important caveat is that mature canopy trees could be as or more important as sprout sources (for broad-leaved species) than as seed sources, and we did not distinguish sprout/seed origin of our regeneration class stems.

Lastly, seedling and canopy richness were similar and generally higher than sapling richness, while saplings had highest evenness, providing mixed support for prediction 1c. It is tempting to interpret similar richness for

seedlings and canopies as an indicator of diversity maintenance into the future, as seedlings become saplings and saplings become canopy trees. However, the actual pool of species in the seedling class with high likelihood to reach the canopy is smaller than the total pool of species present; for example, ironwood is typically a subcanopy species even in maturity, and American beech is likely to face mortality from beech bark disease prior to reaching maturity. Other seedling or sapling species may also be filtered out in later stages due to a combination of environmental factors (Lusk & Laughlin, 2016). Lower SR in the sapling than seedling layer may be circumstantial evidence for this effect. Higher diversity in the canopy than the sapling stage is not inconsistent with this idea. However, most canopies at least partially reflect regeneration drivers operating ~100 years ago when canopy trees were seedlings and saplings (e.g., low deer populations, high light environments following intense harvest disturbance; Henry, 2022), not drivers operating in their understories now (e.g., high deer populations, low light environments from low-intensity harvests). Our results highlight the importance of considering species composition when analyzing metrics of species diversity, particularly for developing management goals or recommendations.

Drivers of individual species regeneration density

Overall, our results supported our hypotheses that local seed sources, germination substrate, canopy density, and deer use will have differential impacts on tree regeneration among species and size classes.

Canopy composition emerged as the most consistent predictor of seedling (and to a lesser extent sapling) composition, offering support for prediction 2a. However, although small-seeded species have previously been found to be less reliant on local seed sources (Ribbens et al., 1994; but see Willis et al., 2016) due to greater seed production potential and longer passive dispersal distances (Burns & Honkala, 1990), we did not identify differential canopy associations between small- and large-seeded species, either with our density models or from patterns of stand-level relative density. Instead, our results indicate that stand-level seed sources are important for both species-specific regeneration (Figure 4) and, collectively, for species-diverse regeneration (Figure 3). Illustrating the lack of seed-size patterns, five species were sampled most commonly only in the seedling class—cherry species, red oak, American elm, white pine, and white ash (Table 4), and these species varied widely in seed size and dispersal strategies. For white ash, this pattern is clearly driven by near-complete mortality of

canopy trees caused by emerald ash borer; a common feature of the other four species is that they are all relatively longer distance dispersers, either by animals (red oak, cherry species; Vander Yacht et al., 2022), wind (American elm; McEuen & Curran, 2004), or both (white pine; Abbott & Quink, 1970; Dovčiak et al., 2005). However, this pattern did not emerge for smallest seeded long-distance dispersing aspen and birch species (e.g., few seedling-only stand samples). This could be due to very low cover of suitable germination substrates (coarse wood, mineral soil, which we address next), low seed rain density, and low light conditions jointly limiting seedlings of these small-seeded (and shade intolerant) species to very low densities such that dense seed rain from local seed sources is required to produce detectable densities of seedlings in a sample. Effects may be detectable at finer spatial resolutions, e.g., if we had analyzed plot-level data rather than stand-level aggregates; for example, in a stand with both canopy trees and regeneration present, small-seeded species may occur in plots without a canopy tree present more commonly than large-seeded species, but that pattern would be obscured when analyzed at the stand level. However, it is important to note that other factors not related to establishment could also contribute to these patterns, including site–soil resource regimes that do not limit seedling establishment but do limit recruitment into sapling and canopy classes or deer browsing pressure (Walters, Farinosi, & Willis, 2020; Webster et al., 2018). Overall, our results indicate retention of canopy seed source is essential for regeneration of most NHF tree species.

Although not supported by individual species seedling density models, patterns of relative density among size classes suggest that lack of suitable substrate may be limiting small-seeded species (prediction 2b). Many small-seeded species, including eastern hemlock, white spruce, birch, and aspen species, had much lower relative density in seedling than canopy classes. This pattern, combined with reported leaf litter substrate limitation for shade-tolerant hemlock (Marx & Walters, 2008; Willis et al., 2016) and spruce (Mori & Mizumachi, 2005) and very high cover of leaf litter substrate in our stands, collectively suggests leaf litter limits seedling densities by limiting seedling establishment for these small-seeded species. However, other factors could contribute to low relative density of seedlings compared with canopy trees, including high seedling mortality in low light conditions for shade-intolerant species such as aspen species and paper birch (Bolton & D'Amato, 2011; Walters, Farinosi, & Willis, 2020; Willis et al., 2016). The lack of significant associations with the substrate in our species density models could be explained by the combination of very low coverage of BMS and coarse wood substrates

(i.e., <3%) in most stands, fewer nonzero data points for less CS (meaning lower statistical power), and our broad seedling height range (0–137 cm tall), can correspond to a broad age range (e.g., stems <1 m tall can persist for >30 years; Marks & Gardescu, 1998) such that substrate characteristics were potentially very different when we measured them than when older seedlings established.

While we found evidence of reduced sapling density in stands with dense canopies (and therefore less understory light), the pattern was found for mid-tolerant and tolerant species but not for the highly shade-intolerant species as we had predicted (2c). Three species (tolerant sugar maple and red maple, and mid-tolerant and cherry species) declined with increasing canopy density in the sapling class, with two (sugar maple and red maple) additional having greater sapling density in stands with higher BA_{SD} (indicative of greater heterogeneity of light availability) (Figure 4). These three species, plus white ash and red oak, also had patterns of lower relative density of saplings versus seedlings (right-skewed and positive parabola; Figure 5). For these species, the combination of shade tolerance and large seeds may confer the ability to establish in leaf-covered understories and persist in the medium term as seedlings in low light understories. However, energy reserves from large seeds and shade tolerance decrease as trees age/increase in size (Kneeshaw et al., 2006; Sendall et al., 2015); diminishing degrees of shade tolerance may not be sufficient to survive and grow into the sapling class, at least not in the high BA, low light environments characterizing STS management regimes in Great Lakes NHFs. While other factors, such as deer (addressed later in *Discussion*), co-limit seedling to sapling transitions in NHFs (Walters, Farinosi, & Willis, 2020), the negative relationships we report between BA and sapling density for red maple, sugar maple, and cherry provide especially compelling evidence that dense shade is at least partially responsible for seedling to sapling recruitment among species.

However, in support of low light limiting shade-intolerant species (prediction 2c), shade-intolerant species were characterized by very low densities of both saplings and seedlings. For quaking aspen, bigtooth aspen, paper birch, and yellow birch, the most common relative density ranking was only canopy trees (i.e., no seedlings or saplings; Table 4). Regeneration patterns of small-seeded species appear to be driven by factors operating earlier in their life histories than those driving seedling–sapling transitions for mid- to shade-tolerant species; that is, on germination and young seedling survival (e.g., germination substrate, high early mortality from shade) rather than the longer term effects of shade on older seedlings and sapling survival. Mechanistically,

carbon starvation is likely contributing to this trend. However, indirect effects may also contribute, including low carbon energy reserves exacerbating resource shortfalls for seedlings by negatively impacting arbuscular mycorrhizal (AM) colonization (Cheng et al., 2005; Koorem et al., 2017; Neuenkamp et al., 2021); BA-affected red maple, cherry species, and sugar maple are all AM-associated species (Bennett et al., 2017).

Lastly, deer browsing pressure was not a significant predictor of species density except for highly browse-preferred sugar maple, providing very limited support for prediction 2d. Compared with models of regeneration density, there was broader circumstantial evidence for deer browsing effects from patterns of relative density among size classes, with seedling to sapling declines for red maple, sugar maple, red oak, and cherry species. For all these species, except cherry (Horsley et al., 2003; Randall & Walters, 2011), previous studies suggest deer contribute to regeneration bottlenecks (sugar maple [Côté et al., 2004; Leak, 2006; Matonis et al., 2011; Neuendorff et al., 2007; Powers & Nagel, 2009], red maple [Horsley et al., 2003], oaks [Dey, 2014], and species groups including maples, cherries, and red oak examined here [Walters et al., 2022]). In contrast, for several species reported less impacted by deer browsing pressure (ironwood, American beech, white pine; Matonis et al., 2011; Nuttle et al., 2013; Walters, Farinosi, & Willis, 2020), saplings were more abundant than seedling classes, suggesting lower impact of deer (or shade) on their transition from seedling to sapling than for maples, oak, and cherry. It is important to note that differences between seedling and sapling relative densities can be driven by both shade and deer, which are impossible to tease apart in our relative density analyses. Both are likely driving regeneration composition and density. However, differences in relative density patterns among size classes for palatable versus less palatable species of similar shade tolerance support evidence of deer impacts on seedling to sapling transitions. It is also important to note that our deer use model is an estimate, with inherent accuracy limitations due to its development from a limited number of sites. Plus, it estimates deer for the year vegetation was sampled, but current regeneration patterns are the integrated product of many years of (likely highly variable) deer browsing pressure. These limitations may partially explain the weaker deer effects compared with other factors, some of which are easier to measure accurately and perhaps less variable over time (e.g., BA).

Implications for future forest composition

Collectively, our analyses suggest that if current limiting factors persist (local seed sources, germination substrates,

light environment, deer populations), future canopy composition may change and diversity may decrease. The most apparent likely future trend is the loss of shade-intolerant species. Already a minor component in our NHF canopies (<0.5% per species), seedlings and saplings of shade-intolerant, small-seeded bigtooth aspen, quaking aspen, and paper birch are sparse. Canopy individuals of these species are likely ~100 years old (Henry, 2022), given most recruited to the canopy in the open, high light conditions following the intense logging in the early 20th century (Dickmann & Leefer, 2016). Near their expected life spans, canopy aspens and paper birch may continue to decline in the canopy and soon become lost as local seed sources. More shade-tolerant but small-seeded yellow birch and hemlock also have very low seedling and sapling densities, primarily due to substrate limitation (Willis et al., 2016) and high palatability to deer (Walters, Farinosi, & Willis, 2020). These species were more common in canopies of pre-European settlement forests of the region than they are currently (Goring et al., 2016), each comprising <1.5% of canopy trees in our stands. In contrast to intolerant species, hemlock and yellow birch are longer lived (200–300 years vs. 100–150 years) and thus could remain minor components in forest overstories for another century or more, barring harvest and/or negative impacts of novel disturbances (e.g., hemlock woolly adelgid range expansion from climate change; Paradis et al., 2008). Mid-tolerant basswood also shows seedling versus canopy relative density patterns similar to hemlock and yellow birch but is larger seeded. Basswood's pattern is consistent with other reports of limited seedling establishment (for unknown reasons), and a reliance on sprouting for regeneration (Brisson et al., 1994) with sprouts highly preferred as deer browse (M. Walters, personal observation).

Several species well represented in the overstory and seedling layer have a lesser dominance but are still moderately abundant in the sapling layer, portending their continuance in the overstory as future cohorts, albeit at potentially reduced absolute and relative density. These species include sugar maple, red maple, cherry species, and red oak. Importantly, over the resource (Figure 1), reductions in their proportional representation coincide with high relative densities of beech and ironwood in the sapling layer; these and other less CS have low potential for canopy recruitment due to genetic size limitations (ironwood, pin cherry) and insect/disease limitations (beech, white ash, elm). For example, current sapling populations are approximately 50% beech + ash + ironwood. Assuming 0% of these stems can become canopy trees, the approximately 35% of current saplings that are sugar maple actually represent 70% of potential canopy recruits. Given the

preponderance of species with little chance to ascend to canopy status presently in the sapling layer, we suggest well-represented species (proportionally) in the sapling layer with the potential to become canopy trees will likely dominate future overstory cohorts given the continuity of current disturbance regimes; i.e., sugar maple and red maple.

In summary, our analyses suggest continued dominance by maples, with reduced diversity via declines and local loss of shade-intolerant and/or small-seeded species, plus basswood. It is important to point out that our analyses are of composition based on species relative densities within and across size classes. The results for absolute density relations with key drivers (canopy density, deer, seed source density) from this study and others (Henry et al., 2021; Walters et al., 2022) indicate low density and stocking of sapling stems may also present a future challenge to sustainable management and resilience of NHF. Eventual dominance is predicated upon successful recruitment beyond the deer browsing zone (Walters et al., 2016), which may be formidable in certain regions; previous analyses of our stands suggest understocking of poletimber in these stands (Walters, Roloff, et al., 2020), which further suggests uncertain canopy recruitment and a potential timber supply gap, making recruitment into the canopy of current advanced regeneration even more critical.

Management implications

In the context of forest management, our results support several potential alternative management strategies aimed at overcoming seed, substrate, and growth/survival (e.g., BA, deer browsing) limitations to establish species-diverse seedling and sapling classes. Given the current paucity of tree species in all strata (and critically in the seed-producing mature/canopy class), active seeding or planting of additional tree species may be necessary to promote greater diversity (Holl & Aide, 2011). Tree species selection for planting could be coupled with assisted migration strategies to further prepare forests for a variety of future climate scenarios (Dumroese et al., 2015). Provisions for exposing more mineral soil during harvest operations and leaving more trees to die and become CWD on the forest floor would help overcome establishment substrate limitations for several species, especially those with small seeds (Marx & Walters, 2008; Willis et al., 2015). Reducing BA more than is done with current partial harvesting systems in stands would result in higher light levels than currently created following harvest and possibly increase sapling recruitment of shade-mid-tolerant and shade-tolerant

species (Danyagri et al., 2019; Gravel et al., 2010; Hupperts et al., 2019, 2020; Knapp et al., 2019; but see Knapp et al., 2021). Finally, implementing alternative strategies to deter deer browsing, such as leaving treetops as natural barriers, may increase regeneration density and recruitment of species already present on the landscape (Grisez, 1960; Hunn, 2007; Smallidge et al., 2021). Integrated strategies including several of the management modifications are likely to be most effective in increasing species diversity in NHFs in the long term.

Another management implication of our results is that the high variability we found across species in seedling versus sapling densities (and ultimately to canopy classes) emphasizes the potential risk in assuming seedling composition reflects sapling composition. This is particularly important for developing meaningful assessments of regeneration success following harvest. Several factors, including shrub competition (Kern et al., 2013; Walters et al., 2016), deer browsing (Bradshaw & Waller, 2016; Côté et al., 2004; Henry et al., 2021; Miller et al., 2023; this paper), and light resource limitation (i.e., canopy BA, Henry et al., 2021; Walters et al., 2022; this paper), may assert strong and variable bottlenecks on seedling to sapling transitions such that tree seedlings are poor predictors of regeneration success (Harris et al., 2022; Walters, Farinosi, & Willis, 2020; Walters, Roloff, et al., 2020). Managers should focus on saplings (taller than 1.5 m and <10 cm dbh) for assessment of regeneration success following harvest or other management activity.

With certain key species potentially in decline, relative density analyses can be used generally to identify the mix of species, desirable and undesirable for management, potentially competing (at least at high overall density) for growing space. Species with right-skewed (increasing relative density with decreasing size class) and negative parabolic size distributions (with saplings having highest relative density) represent strong competitors in regeneration strata. Right-skewed species include red maple, which has been increasing in relative density in recent decades (Fei & Steiner, 2007). Many of the species, which compete with sugar maple for sapling growing space, are negative parabolic, including American beech, which creates well-documented dense thickets (Collin et al., 2017; Elenitsky et al., 2020; Gravel et al., 2011), and ironwood, which is a subcanopy species. Relative density analyses interpreted in the context of stand dynamics and species silvics prove to be a useful method of quantifying potential species and stand trajectories, including identifying undesirable stand trajectories for management such as high proportions of beech and ironwood.

Caveats

Our snapshot-in-time survey is unable to assess trends within size classes over time, and instead relies on assumptions about current patterns and implications for ultimate recruitment; previous studies have suggested comparing size classes from a single survey may not be appropriate for inferring temporal compositional shifts (in the case of tree range shifts, Malis et al., 2016). Follow-up analyses of species diversity dynamics over time would be useful in confirming the temporal component suggested by our relative density analyses, given the long-term maintenance of current compositional drivers over time (e.g., current management regimes, deer use). Nevertheless, our results suggest that current managed NHFs may face long-term challenges with resilience due to low diversity of species without proactive management interventions. Our study suggests intervention may be necessary to promote diverse, resilient stands, which can adapt to changing timber market shifts, pest/pathogen introductions, and climatic changes.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Data (Henry & Walters, 2023) are available from Dryad: <https://doi.org/10.5061/dryad.612jm647z>.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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